DFE-based simulation model for inference of effect size distributions

Model overview

For each locus j:

$$s_{ud,j} \sim p_{\mathrm{DFE}}(s_{ud}), \qquad X_j \sim p_X(x \mid s_{ud,j}), \qquad \beta_{s,j} \sim p_\beta(\beta_s \mid s_{ud,j}),$$

$$\mathrm{SE}_j = (2X_j(1-X_j)n_{\mathrm{eff}})^{-1/2}, \qquad \hat{\beta}_{s,j} \mid \beta_{s,j}, \mathrm{SE}_j \sim N\big(\beta_{s,j}, \mathrm{SE}_j^2\big).$$

Ascertainment indicator (default MAF mode):

$$\mathbf{1}_j = \mathbf{1}\{\min(X_j, 1-X_j) \geq m^*\}.$$

Observed tuple exported to `mutvar infer`:

$$(\texttt{effect\_allele\_frequency}, \texttt{beta}, \texttt{standard\_error})_j = \Big(X_j, \hat{\beta}_{s,j}, \mathrm{SE}_j\Big).$$

Core effect and scaling relationships

Notation:

- $s_{ud,j}$: underdominant selection coefficient,
- $S_{ud,j} = 2N_e s_{ud,j}$: scaled underdominant coefficient,
- $X_j \in (0,1)$: effect-allele frequency,
- $\beta_{s,j}$: latent effect on the β_s axis,
- $\hat{\beta}_{s,j}$: observed effect estimate,
- SE_j: standard error,
- n_{eff}: effective GWAS sample size controlling observation noise.

All beta quantities in this document are on the selection-scaled axis β_s (not raw trait-scale effect units). In the one-dimensional limit, $\beta_s^2 = S_{ud}$, which fixes the scale of both β_s and $\hat{\beta}_s$.

Effect model under stabilizing-selection limits:

High-dimensional (pleiotropic)

$$\beta_{s,j} \mid s_{ud,j} \sim N(0,\ 2N_e s_{ud,j}) = N(0, S_{ud,j}).$$

One-dimensional (single-trait)

$$\beta_{s,j} \mid s_{ud,j} \in \left\{-\sqrt{2N_e s_{ud,j}},\ +\sqrt{2N_e s_{ud,j}}\right\} = \left\{-\sqrt{S_{ud,j}},\ +\sqrt{S_{ud,j}}\right\},$$

with equal sign probability.

Observation model:

$$\mathrm{SE}_j = (2X_j(1-X_j)n_{\mathrm{eff}})^{-1/2}, \qquad \hat{\beta}_{s,j} \mid \beta_{s,j}, \mathrm{SE}_j \sim N(\beta_{s,j}, \mathrm{SE}_j^2).$$

Ascertainment

Default discovery mode is a MAF threshold:

$$\min(X_j, 1-X_j) \geq m^*.$$

Here v_s^* denotes the fixed variance-contribution cutoff used by variance-threshold ascertainment.

Optional alternative ascertainment modes use variance contribution thresholds with

$$v_{s,j} = 2X_j(1-X_j)\beta_{s,j}^2, \qquad \hat{v}_{s,j} = 2X_j(1-X_j)\hat{\beta}_{s,j}^2,$$

and keep loci by either $v_{s,j} > v_s^*$ (truth-side) or $\hat{v}_{s,j} > v_s^*$ (noisy-side).

If variance-threshold ascertainment is used with target p^*, the implied effective GWAS sample size is:

$$n_{\text{eff}} = \frac{2\ \text{erf}^{-1}(1-p^*)^2}{v_s^*}.$$

Mutvar-facing observed output

Observed output is written for direct input to `mutvar infer` with exactly:

- `effect_allele_frequency`,
- `beta`,
- `standard_error`.

At each locus j, the observed columns map to latent quantities as:

- `effect_allele_frequency` $= X_j$,
- `beta` $= \hat{\beta}_{s,j}$,
- `standard_error` $= \text{SE}_j$.

So the exported `beta` column is on the selection-scaled axis β_s. The exported `standard_error` is on that same β_s axis.

Technical generation details

DFE sampling is performed on

$$\ell = \log_{10}(s_{ud}),$$

using an SSD table $\{(\ell_i, f_i)\}_{i=1}^m$ with interpolation-defined density

$$p_\ell(\ell) \propto \text{interp}(\ell; \{\ell_i, f_i\}).$$

Allele frequencies are sampled from an underdominant SFS conditional on s_{ud}. In this implementation, the SFS kernel is parameterized with $2S_{ud}$, where $S_{ud} = 2N_e s_{ud}$:

$$\log \tau(x \mid S_{ud}) = \log\left(\frac{\theta}{x(1-x)}\right) - (2S_{ud})x(1-x) + \text{erf correction}(x, 2S_{ud}).$$